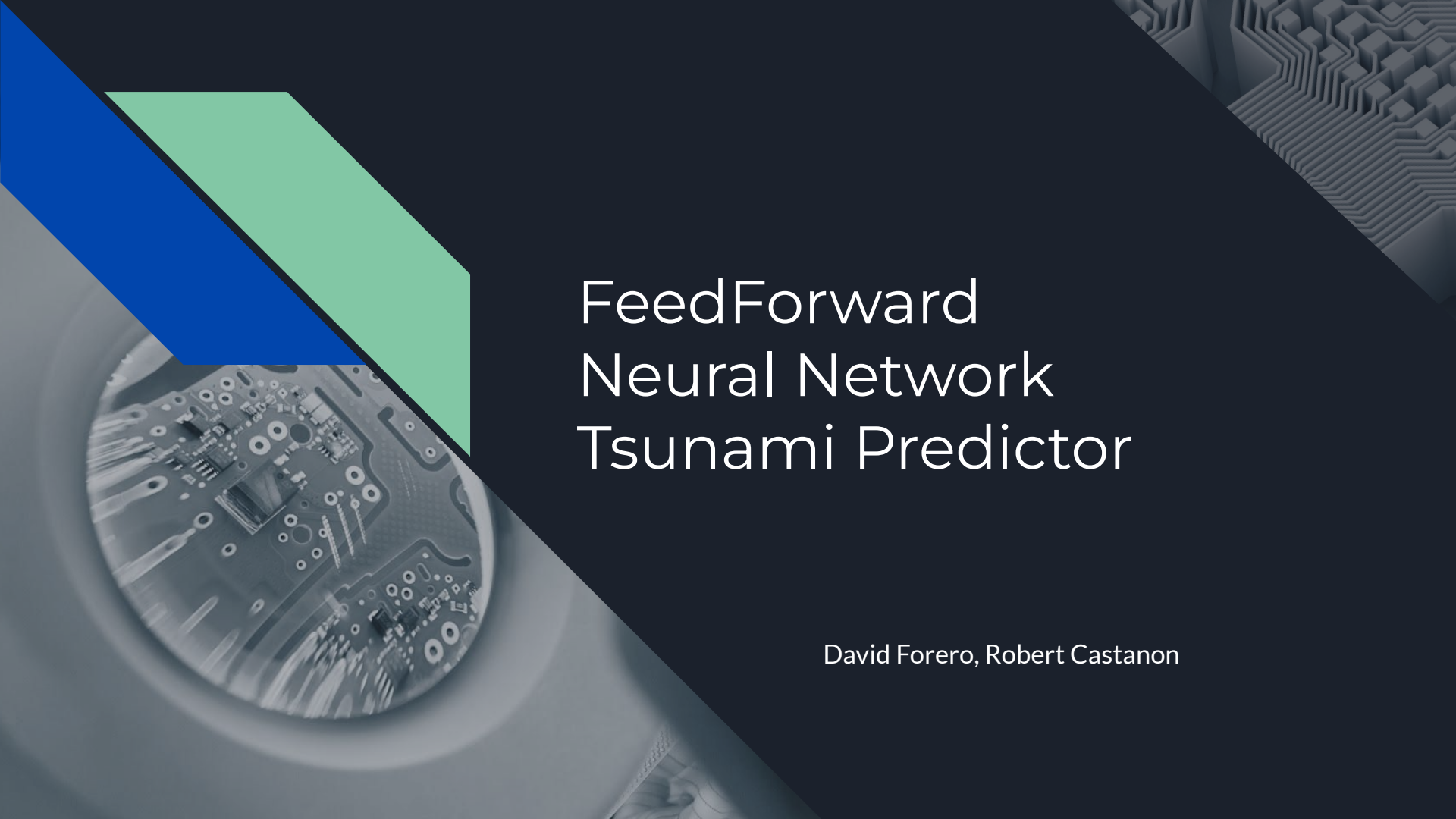


The background is a dark blue gradient. On the left, there is a large, semi-transparent circular image of a circuit board. Overlaid on this and the background are several geometric shapes: a blue parallelogram and a light green parallelogram in the upper left, and a series of white, 3D-looking circuit traces in the upper right.

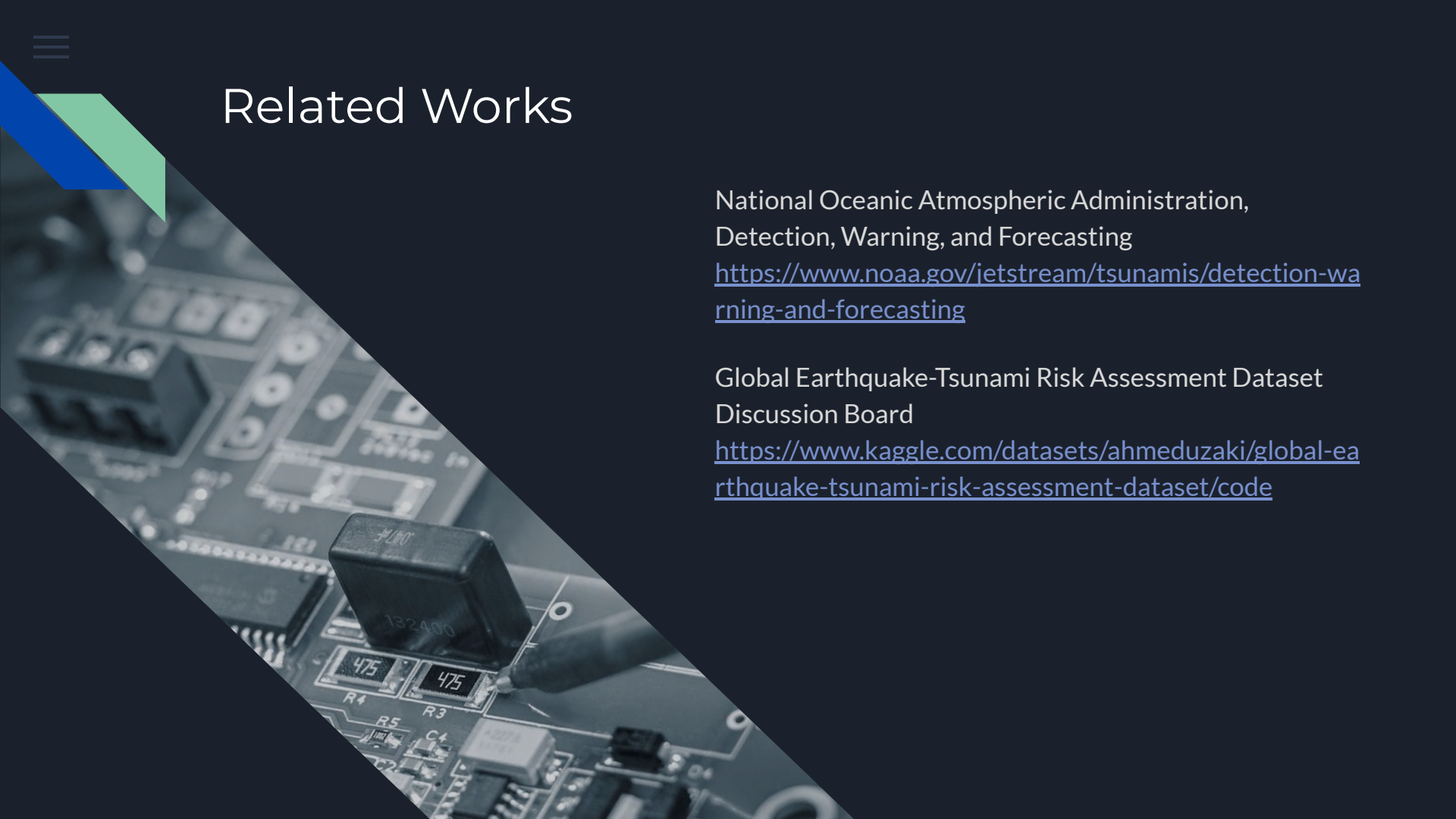
FeedForward Neural Network Tsunami Predictor

David Forero, Robert Castanon



Introduction

Tsunamis cause massive casualties numbers and infrastructure damage every year due to how fast the natural disaster occurs to a naked eye. Tsunamis are formed from unseen underwater earthquakes and using this data with a Neural Network can quickly alert civilians of the danger occurring. Our goal was to create a FeedForward Neural Network with and without Early Stopping to compare the accuracy of the two methods.



Related Works

National Oceanic Atmospheric Administration,
Detection, Warning, and Forecasting

<https://www.noaa.gov/jetstream/tsunamis/detection-warning-and-forecasting>

Global Earthquake-Tsunami Risk Assessment Dataset
Discussion Board

<https://www.kaggle.com/datasets/ahmeduzaki/global-earthquake-tsunami-risk-assessment-dataset/code>



Dataset

Source: Kaggle - Global Earthquake Tsunami Risk Assessment by Ahmed Mohamed Zaki

Type: Structured Numerical

Instances: 782 Instances

Features: Magnitude, Community Determined Intensity, Modified Mercalli Intensity, Significance level, Number of seismic stations, Minimum distance from epicenter, Azimuthal gap, Depth, Latitude, Longitude, Year, Month, Tsunami (binary)

Format: .csv

Pre-processing: 80/20 train-test split with stratification. Standardized to normalize feature scales

Neural Network Methods

- FeedForward Neural Network Model
 - Three hidden layers
 - First = 64 neurons
 - Second = 32 neurons
 - Third = 16 neurons
 - One single-node output layer
 - 1 neuron
- FNN no Early Stopping
 - 300 epochs, 32 batch size
- FNN with Early Stopping
 - Patience = 20
 - Epochs based on validation loss



Evaluation

Metrics

Accuracy

Precision

Recall

F1-score

ROC-AUC

Visualization

Learning Curves

Confusion Matrices





Results

Baseline: Logistic Regression

Accuracy: 0.8598726114649682
Precision: 0.7746478873239436
Recall: 0.9016393442622951
F1 Score: 0.8333333333333334
ROC-AUC: 0.9318647540983606

High Recall, Low Precision

FNN (No Early Stopping)

Accuracy: 0.8726114649681529
Precision: 0.8253968253968254
Recall: 0.8524590163934426
F1 Score: 0.8387096774193549
ROC-AUC: 0.9318647540983606

Better for Non-Linear
Relationships than Baseline

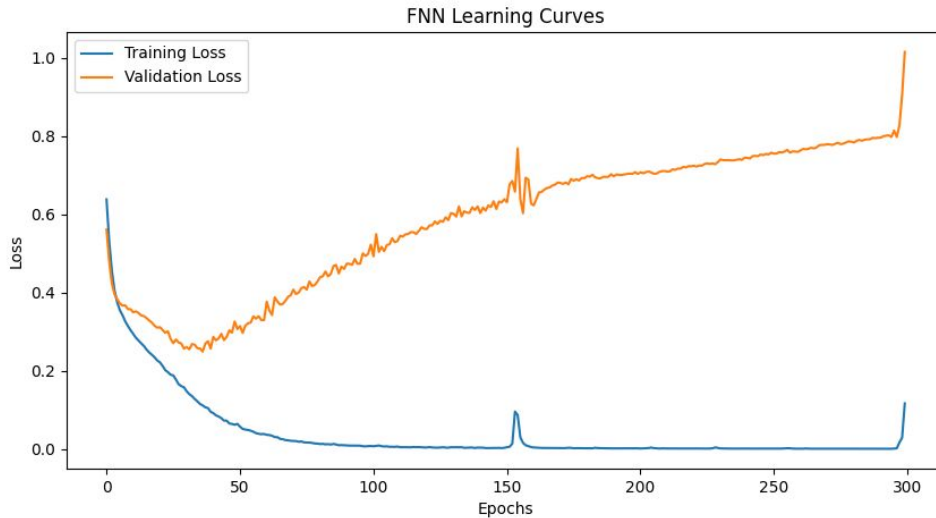
FNN with Early Stopping

Accuracy: 0.9044585987261147
Precision: 0.8285714285714286
Recall: 0.9508196721311475
F1 Score: 0.8854961832061069
ROC-AUC: 0.9465505464480874

Best Performance Overall

Results

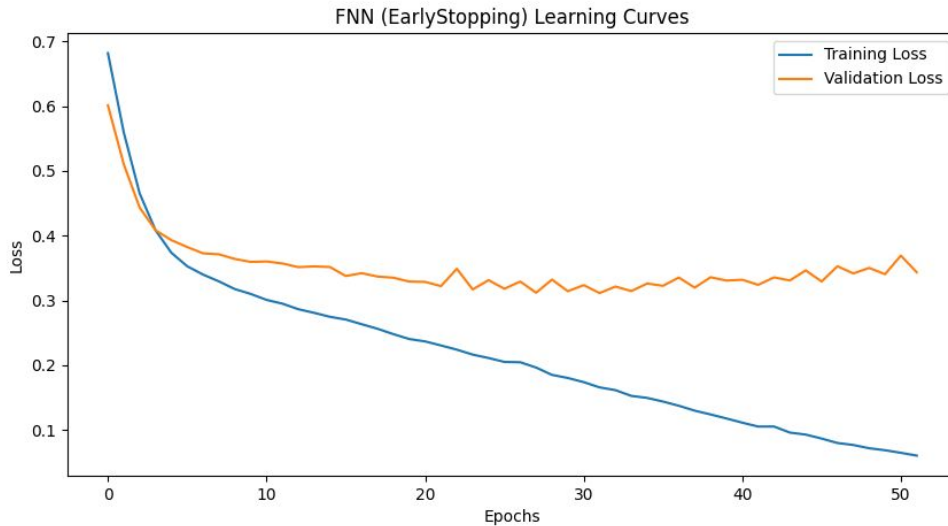
FNN (No Early Stopping)



- Training loss approaches zero
- Validation loss initially decreases
 - Later increases steadily
- Model subject to overfitting

Results

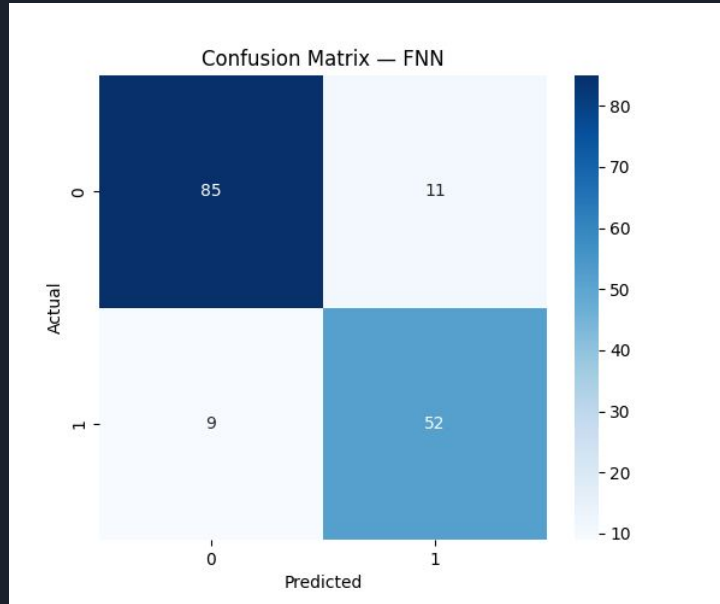
FNN with Early Stopping



- Training loss and validation loss decrease for first ~10 epochs
- Validation loss plateaus and fluctuates
 - Triggers Early Stopping
- Training stops at 50 epochs
- Prevented overfitting for smoother validation loss curve

Results

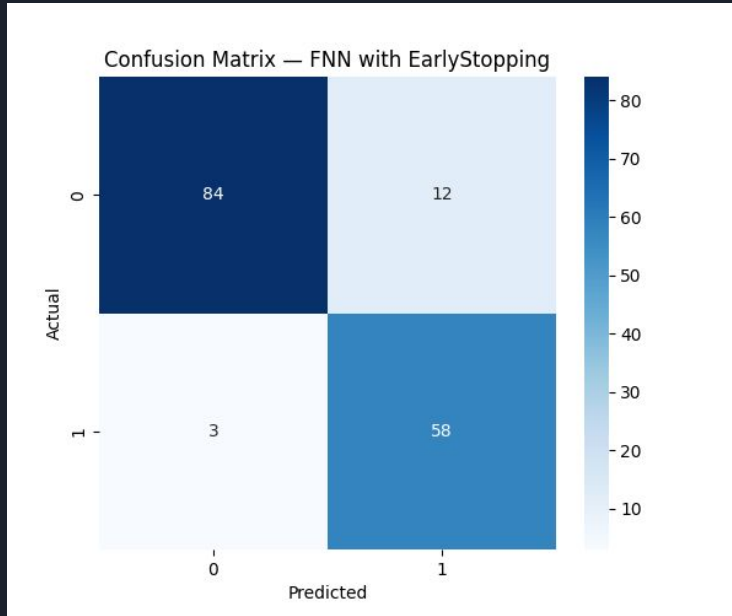
FNN (No Early Stopping)



- 9 false negatives
 - VERY BAD

Results

FNN with Early Stopping



- False negatives reduced from 9 (in FNN) to 3
 - Improved accuracy!
- More reliable for detecting tsunamis
- False positives increased by 1
 - Largely insignificant



Discussion & Analysis

- 
- FeedForward w/o Early Stopping resulted in a higher accuracy but a higher false negative count.
 - FeedForward w/ Early Stopping had a much better generalization of dataset and better track record of identifying real tsunamis.
 - Lack of data and randomness in our model prevent it from producing more accurate results



Teamwork and Task Division

David Forero -

FeedForward Neural Network w/ & w/o Early Stopping, Debugging, Dataset Acquisition, Report Work, Slideshow Preparation

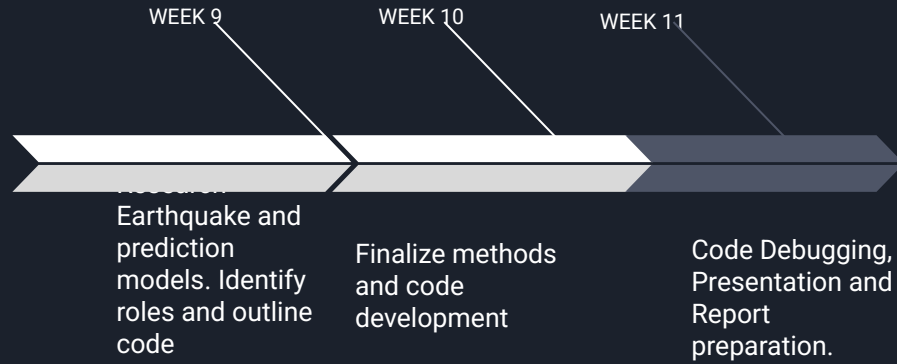
Robert Castanon -

Visualization of Methods, Debugging, Dataset Formatting, Report Work, Slideshow Preparation



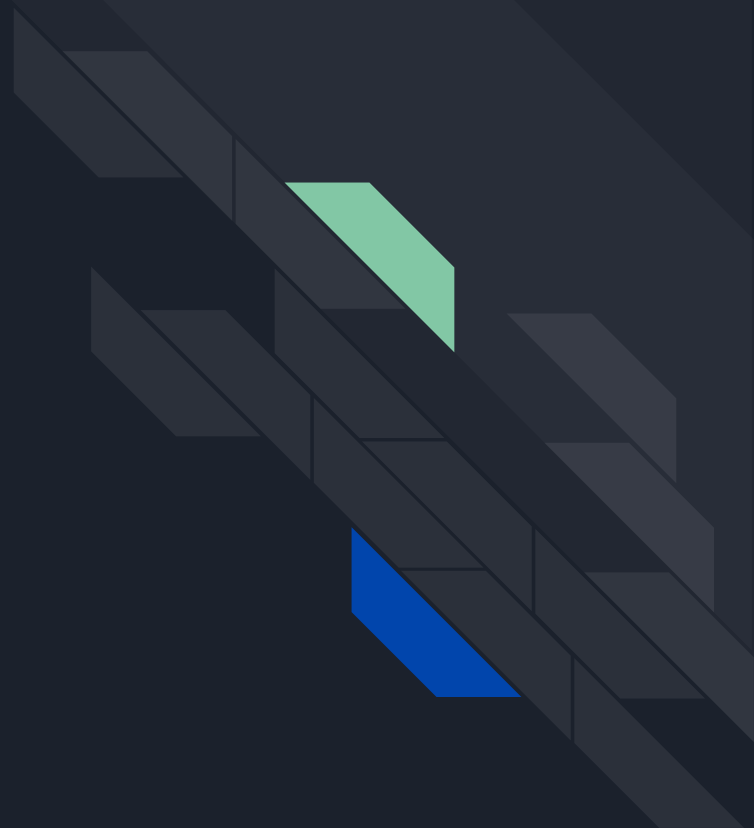


Project timeline



Lessons Learned

- Importance of Data preprocessing
- How much regularization techniques can improve a model
- Why we should compare different methods to each other.
- Importance of verifying a publicly source Dataset carefully and properly





Code Presentation



Thank you!